

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Describe mechatronics system with a neat sketch and give its practical examples.	(CO1)	6
B) Explain Force sensor (Load Cell) with a neat sketch by using strain gauge.	(CO1)	6
C) Discuss the working of AC Motor with a neat sketch and state its advantages.	(CO3)	6
Q.2 Solve Any Two of the following.		12
A) Describe the Data Acquisition System with a block diagram. List any two practical examples.	(CO2)	6
B) Draw a neat diagram of a Seven segment display system? Explain it with an example.	(CO2)	6
C) Explain with a neat sketch On/Off Controller.	(CO6)	6
Q. 3 Solve Any Two of the following.		12
A) Explain the sequencing of pneumatic cylinders using cascade method.	(CO3)	6
B) Give the comparison between pneumatic systems and hydraulic systems.	(CO3)	6
C) Derive model equation of a mechanical system for mass and spring.	(CO5)	6
Q.4 Solve Any Two of the following.		12
A) Explain with a block diagram 8081 microcontroller and its architecture.	(CO4)	6
B) Describe with a neat sketch Programmable Logic Controller (PLC).	(CO4)	6
C) Explain with a sketch of pressure sensor? Give few applications of it.	(CO1)	6
Q. 5 Solve Any Two of the following.		12
A) Explain with a neat sketch of Proportional- Integral -Derivative (PID) Controller.	(CO6)	6
B) Explain the Closed Loop Control System with a suitable example.	(CO5)	6
C) Describe with a neat sketch Proportional- Derivative (PD) Controller	(CO6)	6

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Semester Examination – Summer 2022

Course: B. Tech. Branch : Mechanical Engineering

Semester :VII

Subject Code & Name: BTMEC701 & Mechatronics

Max Marks: 60

Date:13/08/2022

Duration: 3.45 Hrs.

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(Level/CO) Marks

Q. 1 Solve Any Two of the following

- | | | |
|---|-------|-----|
| A) What are the elements of a closed loop control system? | (CO1) | (6) |
| B) What are the various static and dynamic characteristics of a sensor? | (CO1) | (6) |
| C) How a tactile sensor works? | (CO5) | (6) |

Q.2 Solve Any Two of the following

- | | | |
|--|-------|-----|
| A) How a differential amplifier works? | (CO3) | (6) |
| B) What is the specialty of a successive approximations Analog to Digital Converter (ADC)? | (CO2) | (6) |
| C) How a Digital to Analog Converter (DAC) works? | (CO2) | (6) |

Q. 3 Solve Any Two of the following

- | | | |
|--|-------|-----|
| A) Analyse the controlling action of a double-acting cylinder? | (CO6) | (6) |
| B) How the valve bodies are classified? | (CO1) | (6) |
| C) What are the applications of single acting cylinder? | (CO1) | (6) |

Q.4 Solve Any Two of the following

- | | | |
|--|-------|-----|
| A) Explain the microcontroller with a block diagram? | (CO2) | (6) |
| B) What are the various types of registers and their functions in 8085 microprocessor? | (CO5) | (6) |
| C) What do you mean by Ladder Logic? Design a Ladder Logic for a simple task of your choice? | (CO4) | (6) |

Q. 5 Solve Any Two of the following.

- | | | |
|--|-------|-----|
| A) What do you mean by transfer function? Why derivative controller mode cannot be used alone? | (CO6) | (6) |
| B) Explain the response of a first order system to a step-input? | (CO5) | (6) |
| C) What is the specialty of a PID controller? Design a PID controller for temperature control. | (CO6) | (6) |

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	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Define Mechatronics with its merits and demerits. Brief in detail any one of industrial application of Mechatronics.	CO1	6
B) Explain briefly any two of the following transducers: a. Linear-variable-differential transformer (LVDT) b. Optical Encoder c. Strain Gauges	CO1	6
C) Explain Analog to Digital Conversion and Describe any one method of analog-to-digital conversion with neat block diagram.	CO2	6
Q.2 Solve Any Two of the following.		12
A) What is the need for signal conditioning? Explain the steps involved in signal conditioning.	CO2	6
B) Explain with neat diagram DC Motor with its industrial applications in Mechatronic domain.	CO4	6
C) Define Transfer Function and Determine the transfer function $C(s) / R(s)$ of the system shown in Fig.1. by block diagram reduction method.	CO4	6

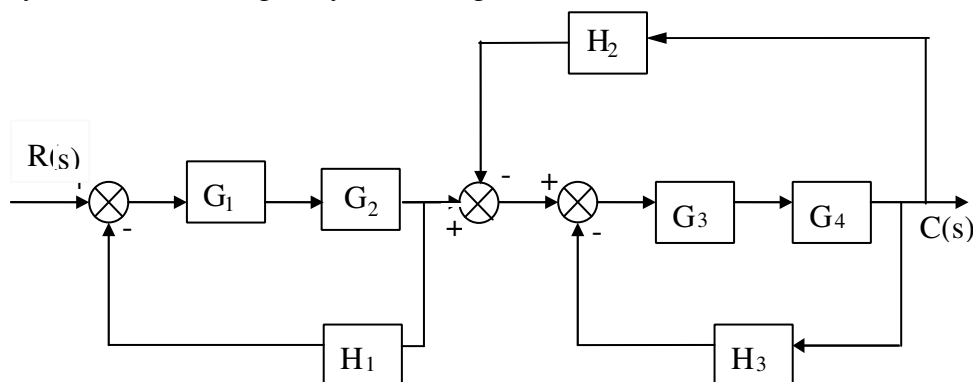


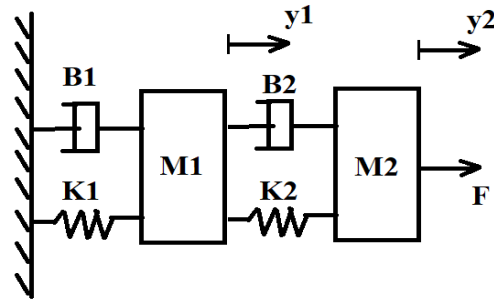
Fig.1.

Q.3 Solve Any Two of the following.		12
A) Explain pneumatic sequencing circuit diagram for 2 cylinders with Cascade method.	CO3	6

- | | | | |
|----|--|-----|---|
| B) | Explain the different types of directional control valves with neat sketches. | CO3 | 6 |
| C) | What is a "Microprocessor"? Explain briefly 8085 microprocessor with the help of a block diagram and also uses of microprocessors? | CO5 | 6 |

Q.4 Solve Any Two of the following. **12**

- | | | | |
|----|--|-----|---|
| A) | Explain Closed Loop Control System with block Diagram. Describe advantages and Disadvantages over Open Loop Control System. | CO3 | 6 |
| B) | Write the differential equations governing the mechanical translational system as shown in figure and determine the transfer function. | CO5 | 6 |



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|----|---|-----|---|
| C) | Explain with block Diagram Programmable Logic Controller (PLC). Write any one industrial application using PLC Programming. | CO4 | 6 |
|----|---|-----|---|

Q. 5 Solve Any Two of the following. **12**

- | | | | |
|----|---|-----|---|
| A) | Explain different control actions used in Closed Loop Control System (PID, PI, PD) and write PID Controllers in brief with advantages. | CO6 | 6 |
| B) | Explain the various Force Sensors with industrial applications. | CO1 | 6 |
| C) | What do you mean by bode plot? Sketch the bode plot for the following transfer function and determine the phase margin & gain margin. $G(S) = \frac{20}{s(1+3s)(1+4s)}$ | CO6 | 6 |

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B.Tech. Branch: Mechanical Engg./Mechanical Engg. (Sandwich)/Production

Subject Code & Name: BTMC701 Mechatronics

Semester :VII

Max Marks: 60

Date:-02-01-2024

Duration: 3 Hr.

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(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Define Mechatronics and list out advantages and disadvantages of mechatronics. Also give some applications	CO1	6
B) Explain with neat sketch working principle of strain gauges. How it can be used as pressure transducer?	CO1	6
C) Explain electromagnetic frequency domain transducer with neat sketch.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain Digital signal with its advantages, disadvantages, and applications.	CO2	6
B) Explain the need for Signal processing.	CO2	6
C) Describe the operational amplifier in detail.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain working principle of stepper motor with neat sketch. Give its classification	CO3	6
B) What are the various types of pressure control valves used in hydraulic systems? Explain any one with neat sketch	CO3	6
C) Explain as to what do you mean by pneumatic air preparation and significance of FRL unit.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Draw and explain the architecture of the 8085 microprocessor.	CO4	6
B) What are the addressing modes of the 8085 microprocessor?	CO5	6
C) What do you mean by microcontroller? Differentiate between microcontroller and microprocessor.	CO5	6
Q. 5 Solve Any Two of the following.		12

- | | | |
|---|------------|----------|
| A) Differentiate between Open loop system and closed loop system giving suitable examples. | CO5 | 6 |
| B) Define transfer function. Explain procedure to determine transfer function of a control system. | CO5 | 6 |
| C) What is a PID Controller? How Does a PID Controller Work? | CO6 | 6 |

***** End *****

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(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) What are the different elements of a Mechatronics system? Explain it with an example.	CO1	6
B) What is the difference between a sensor and transducer? How a tactile sensor works?	CO1	6
C) How a capacitive level sensor works? Explain it with a sketch.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Why signal conditioning is done? Explain the steps involved in it.	CO2	6
B) Explain the working principle of seven segment displays? Give applications of seven segment displays.	CO2	6
C) Why Analog to Digital Conversion (ADC) is required? Explain it with a neat sketch.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) How a stepper motor works? Explain it with neat sketch.	CO3	6
B) How sequencing of pneumatic cylinders is made using cascade method? Explain it with an example.	CO3	6
C) How a solenoid works? Explain it with an application using an electro-pneumatic circuit.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Draw and explain the architecture of the 8085 microprocessor.	CO4	6
B) What are the addressing modes of the 8085 microprocessor?	CO5	6
C) What do you mean by microcontroller? Differentiate between microcontroller and microprocessor.	CO5	6

Q. 5	Solve Any Two of the following.		12
A)	Differentiate between Open loop system and closed loop system giving suitable examples.	CO5	6
B)	Define transfer function. Explain procedure to determine transfer function of a control system.	CO5	6
C)	What is a PID Controller? How Does a PID Controller Work?	CO6	6

***** End *****